

Supersaturated solid solution enhanced biodegradable Zn-Mn alloys prepared by mechanical alloying and selective laser melting

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ARTICLE INFO

Article history:

Received 11 October 2022

Received in revised form 28 January 2023

Accepted 1 February 2023

Available online 3 February 2023

Keywords:

Mechanical alloying

Selective laser melting

Zn-Mn alloy

Supersaturated solid solution

Mechanical properties

ABSTRACT

Zn-Mn alloys have been recognized as promising biodegradable metals due to moderate degradation and good osteogenesis. However, the alloying content of Mn has been severely limited by the low solid solubility (0.8 wt%) in conventional preparation process to avoid the excessive formation of brittle intermetallics. In this work, supersaturated solid solution was achieved in Zn-Mn alloys via mechanical alloying (MA) and selective laser melting (SLM). During MA process, the mixed Zn and Mn powders experienced repeated welding and fracturing, which resulted in refined grains and abundant crystal defects that stored extra energy. This could reduce the energy barrier of interatomic diffusion and induce the rapid diffusion of Mn atoms, leading to supersaturated solid solution with ~3.6 wt% Mn homogeneously dissolved in Zn lattices. Moreover, the rapid solidification process of SLM effectively maintained the metastable supersaturated structure. In addition, the Zn-15Mn alloys prepared by MA and SLM were composed of fine equiaxed grains (~4.34 μm) and high content of high-angle grain boundaries (54.8%). Owing to solution strengthening and grain refinement, the Zn-15Mn alloy prepared by MA and SLM exhibited a compressive yield strength of 178 MPa which was 2.8 times and 1.6 times those of pure Zn and Zn-15Mn alloy without MA process, respectively, as well as a more ductile fracture. Meanwhile, the Zn-Mn alloys exhibited accelerated degradation and comparable cytocompatibility to pure Zn. These findings indicated that the combined process of MA and SLM may be an efficient strategy for the preparation of alloys characterized by supersaturated solid solution and refined grain structure.

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1. Introduction

Zn-Mn alloys have become increasingly attractive as biodegradable metallic materials, attributing to its moderate degradation and good osteogenic ability [1–7]. As an indispensable trace element fulfilling the primary needs of human, Mn can promote the regeneration and deposition of collagen fiber, hence control bone remodeling [8–10]. Jia et al. [11] implanted Zn-Mn alloy into the femoral condyle of rat in order to repair bone defects and observed numerous new bone tissues without noticeable abnormalities. More importantly, Mn alloying can cause the lattice distortion of Zn matrix and increase the resistance of dislocation movement [12], which

provides a good direction for solving the insufficient mechanical strength of Zn, which has been reported for only about 20 MPa far below the requirement for bone implant applications [13,14]. However, the maximum solid solubility of Mn in Zn was reported to be ~0.8 wt% at eutectic temperature and became negligible after cooling to room temperature [15,16]. Hence, the alloying content of Mn has been severely limited in conventional preparation process, because higher Mn contents inevitably result in the formation of a large amount of brittle intermetallics, which would on the contrary deteriorate the mechanical properties of Zn-Mn implants [17].

Mechanical alloying (MA) is a highly non-equilibrium preparative technique that can artificially provide the thermodynamic driving force and inter-diffusion channel between different atoms at solid state [18–20]. During MA process, the mixed powders undergo repeated welding, fracturing and re-welding process, which is beneficial for the diffusion of atoms and quick homogenization [21,22].

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Therefore, MA has been widely used in the preparation of various metastable structures, such as supersaturated solid solution [23,24], amorphous structure [25,26], etc. A recent study showed that MA improved the solid solubility of insoluble Nb in Cu matrix to 10.0 wt % (0.06 wt% in equilibrium), resulting in Cu-Nb supersaturated solid solution [27] and thereby enhancing the mechanical strength. Hopefully, MA may also qualify in extending the solid solubility of Mn in Zn matrix, thereby maximally enhancing the mechanical performances of Zn-Mn alloys.

It is worth noting that MA technique mainly focuses on powder materials, which need to be consolidated into bulk structures by subsequent process. However, conventional consolidation techniques are characterized by high pressure and extended period of exposure to elevated temperature [28], which easily result in the loss of the metastable structures [29] formed during MA in the ultimately consolidated parts thereby affecting the mechanical strength of the consolidated alloys. For example, Prosviryakov et al. [30] achieved a Zr content of 20 wt% in Al-based supersaturated solid solution by MA. However, the precipitation of consistent Al_3Zr phase formed during the hot pressing and subsequent annealing process because of the exposure to the pressing temperature for a long period. Therefore, a subsequent rapid consolidating process is definitely needed to retain the metastable supersaturated solid solution structure formed by MA process, hence ensuring the mechanical enhancing effect on the consolidated alloys.

Selective laser melting (SLM) is an eligible technique for the consolidation of alloys with metastable structures [31–35]. This process is characterized by localized high heat input during short interaction time and a rapid cooling rate up to $10^5\sim 10^7$ K/s [36–39]. Theoretically, such nonequilibrium rapid solidification process can facilitate the interception of specific metastable phases while endowing as-processed alloys with fine microstructure [40–42]. Chen et al. [43] successfully prepared a high-strength Ti–37 Nb–6Sn alloy with metastable β -type resulted from the effect of rapid solidification of SLM. Kharanzhevskiy et al. [44] prepared a Fe–Cu solid solution alloy by SLM, the Fe–Cu supersaturated structure formed during MA process was retained, since the high velocity of the movement of the front led to the nonequilibrium capture of the impurity during SLM rapid cooling, thus the structure had no time to relax to the equilibrium state, which led to the formation of supersaturated Fe–Cu solid solution. Therefore, SLM is expected to be a potential consolidating technique in preventing the precipitation of the supersaturated solute element resulting from MA process, thereby preparing Zn–Mn alloys with metastable supersaturated solid solution hence enhancing the mechanical strength of Zn–Mn alloys. To our knowledge, correlative studies have been barely reported.

In this work, a combined strategy of MA and SLM was proposed for the first time to prepare supersaturated solid solution strengthened Zn–Mn alloys. The evolution of microstructure and solid solubility of MAed Zn–Mn powders and subsequent SLMed Zn–Mn alloys with different Mn contents were systematically investigated. And particular attention was paid to exploring the mechanisms of solid solubility extension. Moreover, the mechanical properties of the resulting Zn–Mn alloys were studied and the corresponding reinforcing mechanisms were discussed in detail on the basis of solid solubility, grain sizes, grain boundary angles and fracture modes. Furthermore, the degradation properties and cytocompatibility of the Zn–Mn alloys were also assessed for possible biodegradable implant applications.

2. Material and methods

2.1. MA process

Pure Zn powders (purity 99.9%, Shanghai Naiou technology Co., Ltd.) of 75 μm average particle size and Mn powders (purity 99.9%,

Shanghai Naiou technology Co., Ltd.) with particle sizes of 2–10 μm , were used for the MA process, which was carried out in a planetary ball milling machine (Pulverisette 6, Fritsch, Germany). The Zn and Mn powders were mixed and ball-milled at a rotation speed of 300 rpm for 15 h in an argon atmosphere, tungsten Carbide vial and balls of 10 mm and 6 mm diameters were selected as milling media, where the ball-to-powder ratio was 20:1, as these parameters led to uniform size and proper shape of particles for subsequent SLM process after checking by a series of experiments. The ball milling process was cyclic, and the milling mode was set to a 6 min suspension between each run for 10 min. And 2 wt% anhydrous ethanol was added into the vial as process control agent (PCA) to prevent excess cold welding as well as powder oxidation. After MA, the powders were sieved to 15–53 μm using wire sieves of 900 and 270 mesh and a vibratory shaker. The MAed Zn–xMn ($x = 5, 10, 15$ and 20 wt%) alloy powders were eventually obtained for the subsequent SLM process. Physical mixing process was also carried out in a planetary ball milling machine (Pulverisette 6, Fritsch, Germany). The Zn and Mn powders were mixed and ball-milled at a rotation speed of 100 rpm for 0.5 h in an argon atmosphere. Tungsten Carbide vial and balls of 6 mm diameters were selected as milling media, where the ball-to-powder ratio was 5:1. The ball milling process was cyclic, and the milling mode was set to a 5 min suspension between each run for 10 min

2.2. SLM consolidation

The MAed powders were consolidated into Zn–xMn alloys by utilizing a self-developed SLM equipment, which mainly consisted of computer control module, gas protection system, scanning galvanometer coupled with fiber laser with maximum power of 500 W and wavelength of 1064 nm. Before consolidation, argon gas was filled into the building chamber to keep an almost oxygen-free atmosphere with oxygen content below 100 ppm and avoid oxidation during the SLM process. The laser power, scanning speed and scanning line spacing parameters used for SLM 3D printing process in this study were carefully chosen after a series of pre-experiments and modifications, in order to ensure the optimum consolidating quality of the SLMed alloys. All SLM parameters were determined as follows: laser power 70 W, hatch spacing 80 μm , scanning speed 70 mm/s, raster angle 90°, laser spot diameter 50 μm , layer thickness 100 μm . In addition, pure Zn and Zn–15Mn alloy without MA process were also prepared by SLM as control groups.

2.3. Microstructure characterization

The particle size distribution of the MAed Zn–xMn powders was obtained using a laser particle size analyzer (Mastersizer 3000, Malvern Instruments Co., Ltd., UK). The morphologies and elemental distribution of representative MAed Zn–xMn particles were studied using scanning electron microscopy (SEM, Phenom ProX, Phenom, Netherlands) with energy dispersive spectrometer (EDS). And transmission electron microscopy (TEM, Tecnai G2 F20, FEI, USA) was also employed to study the microstructure of the MAed powders. Moreover, a fast Fourier transform (FFT) and inverse-FFT were conducted on the high-resolution TEM (HRTEM) micrographs to evaluate the lattice structure.

The phase compositions of MAed Zn–xMn powders and SLMed Zn–xMn alloys were detected using X-ray diffraction (XRD, D8 Advance, Bruker Co., Ltd., Germany) with a scanning speed of 5 °/min. Furthermore, the solid solubility (at%) of Mn in Zn after MA was calculated by Vegard formula (1), which had been commonly used for quantitative calculation [45,46].

$$(1 - n)a_1 + na_2 = a \quad (1)$$

where a_1 and a_2 are the lattice parameters of the solvent element and solute element, respectively, a refers to the lattice parameter of the MAed Zn-xMn powders and n is the solid solubility (at%) of Mn in the MAed powders. All the lattice parameters of as-received and MAed powders were calculated by digital analyzing software MDI Jade 6.5.

For microstructural observation, the Zn-xMn alloy samples were prepared after which SEM and EDS were applied to study the structure and composition of the SLMed alloys. In addition, differential scanning calorimetry (DSC) analysis was performed utilizing a synchronous thermal analyzer (STA-200, Nanjing Dazhan Institute, China) to determine the melting temperature and solidification range of the SLMed Zn-xMn alloys. And the data were recorded from 0° to 600°C at a heating rate of 15 °C/min under argon atmosphere.

2.4. Mechanical properties

To investigate the compressive performance of the prepared Zn-xMn alloys, compression tests were proceeded. And cylinder samples with 3 mm diameter and 4.5 mm height were hard-pressed using an universal tester (ZLC-50 M, Jinan Zhongluchang Testing Machine Manufacturing Co., Ltd., China) at a loading speed of 0.2 mm/min. Thereafter, to determine the specific fracture modes, the fracture morphology of the Zn-xMn alloys was characterized by SEM. The micro-hardness of the Zn-xMn alloys was also measured by a Vickers hardness tester (HXD-1000TM/LCD, Shanghai Taiming Optical Instrument Co., Ltd., China) with a load of 4.9 N for a dwelling time of 15 s. And the average hardness values were calculated from ten measurements for each composition.

2.5. Electrochemical tests

To evaluate the degradation properties of the Zn-xMn alloys prepared via MA and SLM combined process, electrochemical tests (ASTM G102-89) including open-circuit potential (OCP), potentiodynamic polarization (PDP) and electrochemical impedance spectroscopy (EIS), were carried out in simulated body fluid (SBF) solution, which could provide a similar liquid environment to human body fluids. A standard three-electrode system was utilized with a saturated calomel electrode (SCE) as the reference electrode, a platinum sheet as the auxiliary electrode, and the Zn-xMn alloy samples determined as working electrode. Before PDP measurements, the Zn-xMn alloy samples were pre-immersed in SBF solution for 3600 s to reach a stable state. Afterwards, EIS tests were performed at OCP from 0.01 Hz up to 10^4 Hz with amplitude of ± 10 mV. Finally, PDP measurement was implemented at a scan rate of 1 mV/s, starting from a potential 500 mV below the OCP. The corrosion potential (E_{corr}) and corrosion current density (I_{corr}) were extracted from the PDP curves by Tafel extrapolation.

2.6. Cytocompatibility tests

Cell fluorescence test and cell viability test are considered to be the most common methods in evaluation the biocompatibility of biodegradable implants [47,48]. Human osteosarcoma cells (MG-63) were used to investigate the cytocompatibility of the prepared Zn-xMn alloys. The extract mediums was prepared according to ISO 10993-5/12 and ISO 23317 [49-53]. The alloy samples were cleaned, sterilized, and incubated in Dulbecco's modified Eagle's medium (DMEM, Gibco, USA) with 10% fetal bovine serum for 72 h under physiological conditions (5% CO₂, 20% O₂, 95% humidity and 37 °C) with an extraction ratio of 0.2 g/ml (mass/volume). MG-63 cells with a density of 5×10^3 cells per well were seeded into a 96-well culture plate with DMEM and incubated for 1 d to allow for attachment. Thereafter, the DMEM medium was replaced by the alloy extracts with concentrations of 50% and 10% then cultured for 1, 3 and 5 days,

respectively. For cell viability, CCK-8 solution was utilized to each well then incubated for another 2 h. Paradigm Detection Platform (Beckman Coulter, USA) was used to measure the optical density at a wavelength of 450 nm, which was positively correlated with cell number. To obtain the fluorescence morphology, the cells were tenderly rinsed with phosphate buffered saline solution and then stained with Calcein-AM and Ethidium homodimer-1 reagents for 15 min at 37 °C. Subsequently, fluorescence microscopy (BX60, Olympus, Japan) was used to observe the cells which were mounted onto glass slides.

2.7. Statistical analysis

In this work, all experimental results were presented as mean with standard deviation (mean $\pm$ SD) from at least 3 replicates, and analyzed using SPSS software with p value < 0.05 (*) to reveal a significant difference between the data by analysis of variance (ANOVA).

3. Results and discussion

3.1. Microstructure characteristics of Zn-xMn alloys

XRD patterns of the as-received Zn and Mn powder, MAed Zn-xMn powders and the physical-mixed Zn and Mn powder were displayed in Fig. 1. As could be seen in Fig. 1a and b, the as-received powder samples were consisted of diffraction peaks of Zn and Mn, respectively, while no other phases were detected. These results were consistent with the standard XRD pattern of Zn and Mn, respectively, indicating the high purity of the as-received powders. All the diffraction peaks were indexed according to PDF#04-0831 (Zn) and PDF#32-0637 (Mn), including 002_{Zn}, 100_{Zn}, 411_{Mn}, 101_{Zn}, 332_{Mn}, 102_{Zn}, 103_{Zn} and 004_{Zn}. And the XRD patterns of the MAed powders showed obvious changes in the powder microstructure during the milling process. After MA, the diffraction peaks of Zn not only showed a remarkable broadening with an intensity reduction but also shifted to lower angles, whereas the Mn peaks almost disappeared, as shown in Fig. 1c. This might be ascribed to the distortion of Zn lattice structure resulted from the solid solution of Mn and the larger atomic radius of Mn than Zn. According to Bragg's Law, the diffraction peaks of Zn would shift to lower angles. This phenomenon was consistent with the rules reported by Lei et al., who found similar changes in the diffraction peaks of Cu-Nb powders after MA process [27]. The shift of diffraction peaks revealed the formation of Zn-Mn supersaturated solid solution during the MA process. Moreover, the diffraction peaks gradually shifted towards much lower angles as Mn content increased, which was more evident for the peaks of 101_{Zn} and 100_{Zn} in the enlarged patterns (Fig. 1d,e). And the peak shifting reached a plateau when Mn content increased to 15 wt % with no indistinguishable shifting compared with the pattern of MAed Zn-20Mn powder. This might also indicated a maximal extension of solid solution in MAed Zn-15Mn and Zn-20Mn powders.

To evaluate the efficacy of MA process on Zn-Mn solid solution more rigorously, quantitative analysis towards the solid solubility (wt%) of the MAed Zn-xMn powders was further conducted based on the XRD data. According to Vegard formula, the solid solubility were determined to be 0.4, 0.8, 3.6, and 4.0 wt% for Mn contents of 5, 10, 15, and 20 wt%, respectively. The variation of solid solubility showed a consistent trend with the peak shifting, i.e. at lower Mn contents, Mn atoms were rapidly dissolved into the matrix and induced the continuous formation of Zn-Mn supersaturated solid solution during MA process, whereas the dissolution ratio became saturated for Mn contents of 15 and 20 wt%. The extension of solid solubility of the MAed Zn-Mn powders could be resulted from the following aspects. On the one hand, MA process increased crystal defect density and microstress inside the powders, which was conducive to reduce the

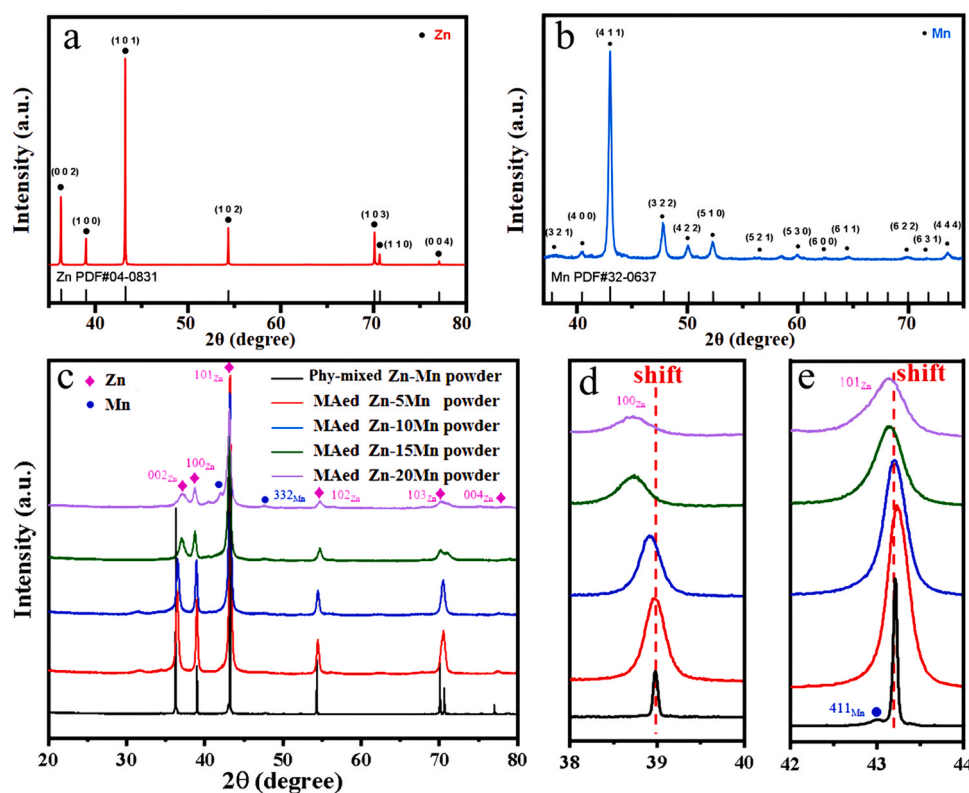


Fig. 1. XRD patterns of the as-received powders, physical-mixed Zn and Mn powder, and MAed Zn-xMn powders.

effective reaction energy and facilitate solid reaction propagation. On the other hand, the large amount of external energy gathering at the crystal defects would lead to increased chemical activity of powder particles, decreased energy barrier for atomic diffusion, and rapid diffusion of Mn atoms. In addition, the free energy of Zn-Mn system increased on account of the increase of crystal defects during MA process, while the solution of Mn could promote the solid solution process. The extension of solid solubility to 3.6 and 4.0 wt% in the present study indicated the high efficacy of MA process on preparing Zn-Mn supersaturated solid solution, which was beneficial for the mechanical performance.

In order to achieve an intuitive understanding about the crystal structure changes induced by MA process, the MAed Zn-15Mn powder was further representatively characterized by TEM (Fig. 2), as the extension of solid solubility was close to saturation when Mn content reached 15 wt%. The low-magnification morphology of a typical MAed Zn-15Mn particle in Fig. 2a showed a uniform and integrated structure. A close observation towards the microstructure was carried out by HRTEM (Fig. 2b), which displayed two typical regions A and B with different lattice structure. The corresponding SAED pattern was given as an inset in Fig. 2b which showed a combination of Debye-Scherrer rings with bright spots, indicating a polycrystalline structure of the MAed Zn-15Mn powder. Moreover, the indexed pattern was composed of the diffracted planes of Zn (102), Zn (002), Zn (110), Zn (101), and Mn (332) in accordance with the XRD results. To further study the detailed lattice structure, FFT and IFFT analysis was conducted in the two regions A and B, as presented in Fig. 2c,d, respectively. It was evident from Fig. 2c that the interplanar spacing in region A was about 0.274 nm, larger than the standard value of Zn lattice, which could be ascribed to the dissolution of Mn. This further led to the distortion of Zn lattice as a result of the relatively larger atom diameter of Mn, which was consistent with the XRD results. As a result, large amounts of lattice distortions could be observed in the IFFT pattern, which was believed to increase the resistance of dislocation movement.

According to the FFT and IFFT analysis shown in Fig. 2d, the atoms were arranged in a bcc structure with an interplanar spacing of about 0.190 nm, which corresponded to Mn (332) of undissolved Mn phase. These findings indicated that the MA process effectively extended the solid solubility of Mn and caused large amounts of lattice distortions in the MAed Zn-Mn powders, and the undissolved Mn existed as fine-dispersed phases in Zn matrix.

To study the morphological changes of Zn-xMn powders after MA process, the physical-mixed powder and MAed Zn-xMn powders were further observed by SEM and the results were shown in Fig. 3. During the milling process, the powder particles suffered from a seriously repeated process of deformation, cold welding and fragmentation, leading to continuous change of shape and size. As shown in Fig. 3a, the physical-mixed Zn-Mn powder contained Zn particles with a smooth surface and good sphericity. This implied that no severe deformation or damages were induced on Zn particles after the short-time and low-energy collision between grinding balls and powder particles in the physical-mixing process. In addition, it could be observed that the flake-like Mn particles were uniformly distributed among Zn particles in the physical-mixed powder. After MA, the powder particles with low Mn content (5 wt%) exhibited polygonal appearance but relatively uniform size distribution (Fig. 3b). This demonstrated that the powders were subjected to intense impact from the grinding balls during the high-energy milling process, thereby resulting in the deformation of powder particles. When Mn content increased, the particle size of MAed Zn-10Mn powder increased obviously accompanied by many fine particles and flake particles, as shown in the SEM micrograph in Fig. 3c. At relatively higher Mn content of 15 and 20 wt%, the particle size of MAed Zn-15Mn and Zn-20Mn powders further increased. It could be seen from Fig. 3d,e that most of the powder particles presented bulk shape with a certain sphericity and rough surfaces, which indicated that the powders underwent repeated welding and fragmentation process.

The particle size distribution of the physical-mixed powder and MAed Zn-xMn powders was also measured and displayed in Fig. 3.

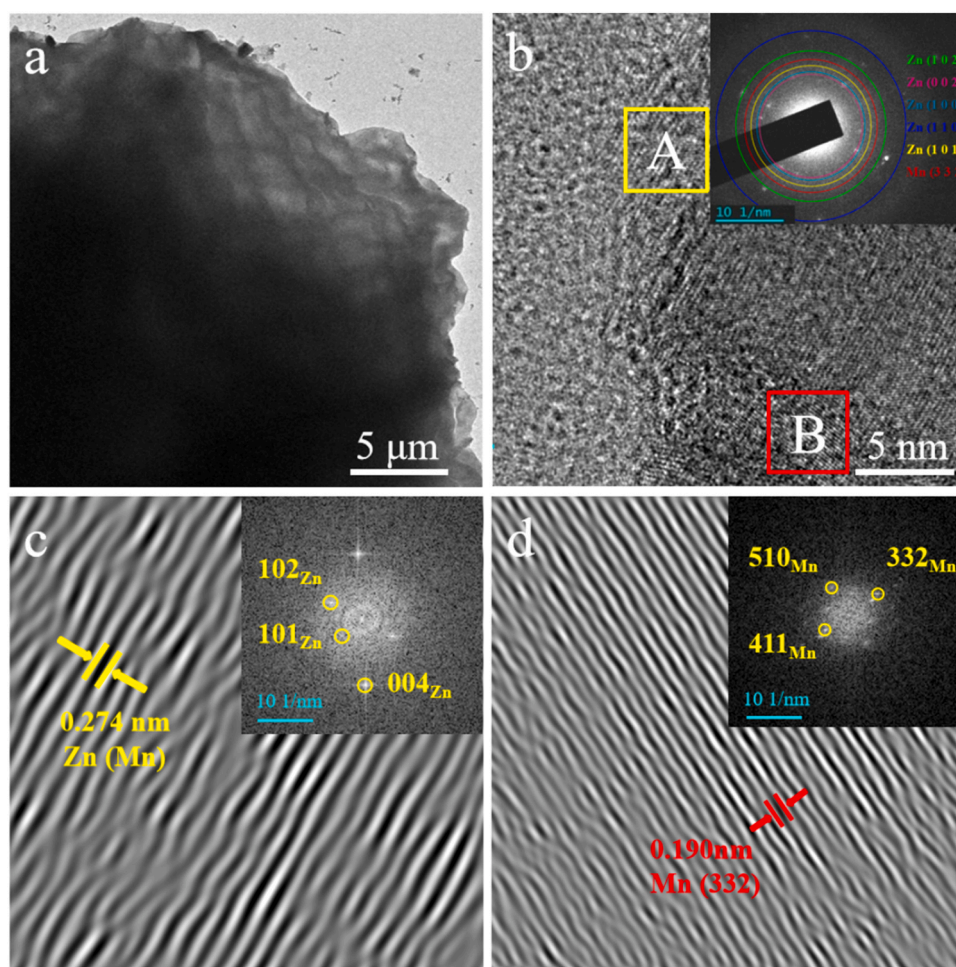


Fig. 2. (a) TEM, (b) HRTEM micrographs of MAed Zn-15Mn powder. The FFT and IFFT analysis at (c) region A and (d) region B indicated the Zn-Mn supersaturated solid solution and undissolved Mn, respectively.

The physical-mixed powder showed a specific bimodal particle size distribution with the median particle sizes of 72.14 μm and 8.64 μm , respectively. In comparison, powders after MA showed a unimodal peak mode of particle size distribution with approximately logarithmic normal distribution and refined sizes. During the MA process, high energy absorbed from the collision between grinding balls and powder particles resulted in a large amount of deformation in large Zn particles, after which the large particles broken into many tiny particles due to the fact that Zn particles are more brittle compared with Mn particles. Tiny broken Zn particles could be embedded with soft Mn particles due to the welding phenomenon hence forming larger alloy particles, which might contributed to the formation of Zn-Mn supersaturated solid solution. As the Mn content increased, the measured $D_v(50)$ value of the MAed powders firstly increased from 14.46 μm to 32.21 μm , followed by a decrease to 22.66 μm . The morphology and size distribution results illustrated that MA process could achieve the effective integration of Zn and Mn powders while significantly refining the powder particles by the high energy impact between the powders and grinding balls/vial.

To reveal the alloying homogeneity inside MAed powders, the cross-sectional microstructure and element distribution of the MAed Zn-xMn powder particles were analyzed by SEM/EDS and shown in Fig. 4. SEM micrographs showed that physical-mixed powder was consisted of both circular and polygonal particles with dense microstructure, which were determined as interspersed Zn and Mn particles, respectively, by EDS mapping. After 15 h of MA process, the cross section of the particles still showed satisfying quality and

complete structure with some tiny dark spots dispersed inside the particles, as marked by the yellow circles. Since back scatter signals are correlated to the atomic number of elements, the bright and dark areas in cross-sectional SEM micrographs corresponded to different compositions. According to the EDS mapping results, the powder particles were dominated by Zn element, as well as uniformly distributed Mn element which almost coincided exactly with Zn element, indicating the well dissolution of Mn into the Zn matrix after MA process. Moreover, EDS mapping also showed that the tiny dark spots in the SEM micrographs were enriched zones of Mn element where Zn element was absent, which was more obvious with at higher Mn contents. This indicated that the undissolved Mn element was distributed inside the particles as dispersive phases, which was consistent with the TEM results. It was speculated that the welding between powders would wrap the undissolved Mn inside the powder particles during the MA process, and the embedding phenomenon of Mn in the cross section of the powder particles became more noticeable with the increase of Mn content due to the solid solubility limit. In a word, after the MA process, Mn element was either successfully dissolved into Zn matrix forming supersaturated solid solution or embedded in the matrix as dispersive phases.

In order to investigate the effect of consolidating process on the metastable Zn-Mn powders, XRD analysis was performed on the SLMed Zn-xMn bulk alloys (Fig. 5a). As could be seen, the 101_{Zn} diffraction peak of SLMed Zn-xMn alloys was comparable to that of MAed Zn-xMn powders, respectively, despite very slight shifting to higher angles. This indicated the effect of SLM process in preventing

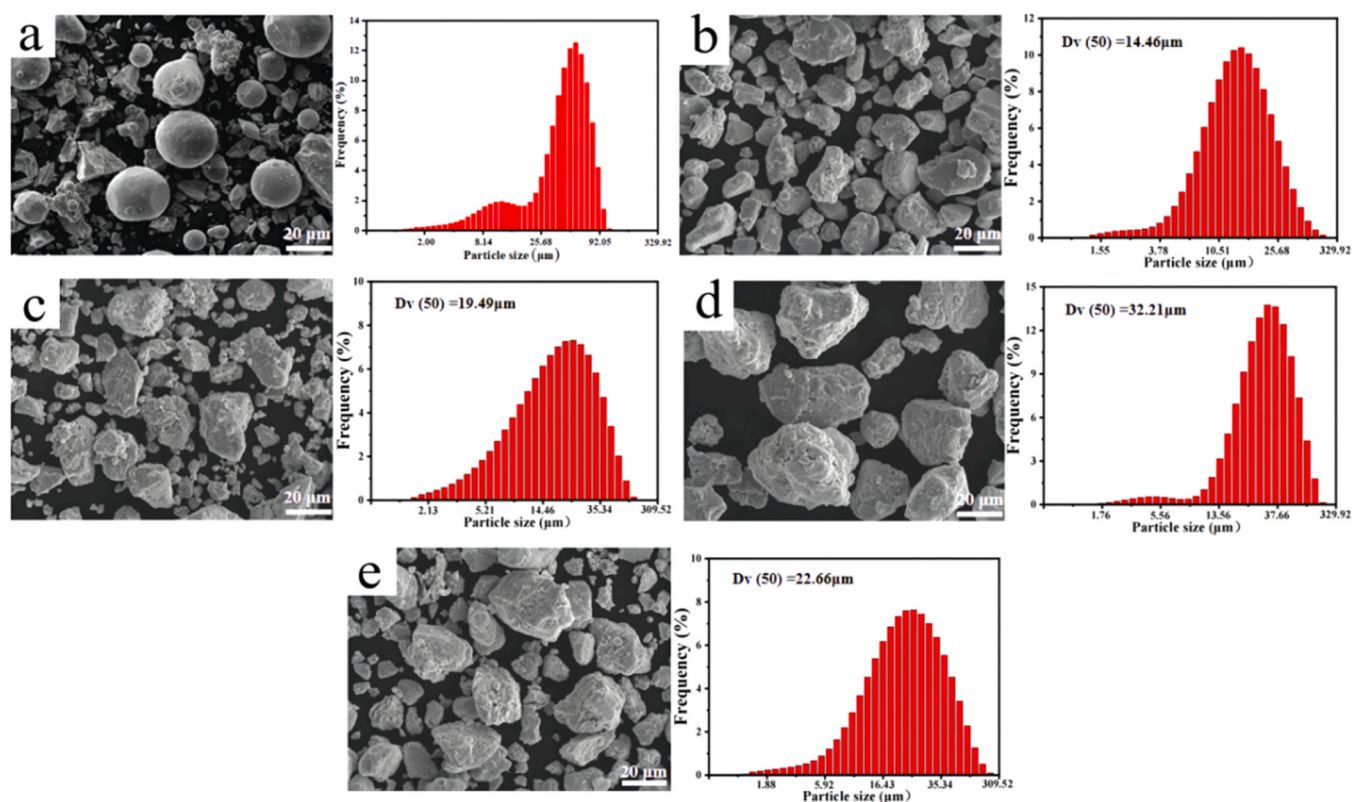


Fig. 3. SEM micrographs and size distribution of (a) physical-mixed powder and MAed (b) Zn-5Mn, (c) Zn-10Mn, (d) Zn-15Mn and (e) Zn-20Mn powders.

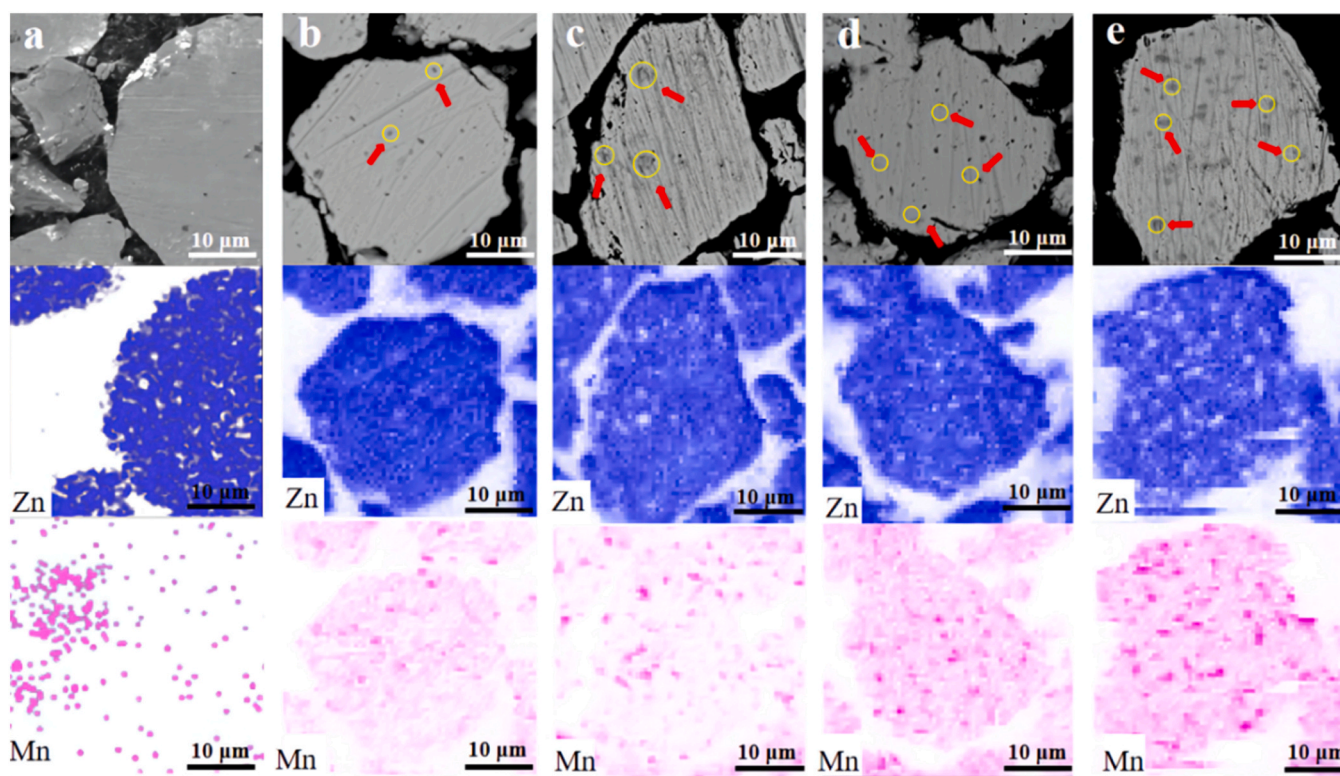


Fig. 4. SEM morphologies and EDS elemental mapping of the (a) physical-mixed powder and MAed Zn-xMn powders: (b) Zn-5Mn, (c) Zn-10Mn, (d) Zn-15Mn and (e) Zn-20Mn.

the precipitation of the supersaturated solute Mn originated from MA process, thereby maintaining the metastable supersaturated structure in the Zn-Mn alloys. As can be seen in Fig. 5a that with

regard to Zn-5Mn and Zn-10Mn alloys, the major peak 101_{Zn} shifted to higher angle for 0.23% and 0.52%, compared with the corresponding MAed powders. Which indicated that there could be

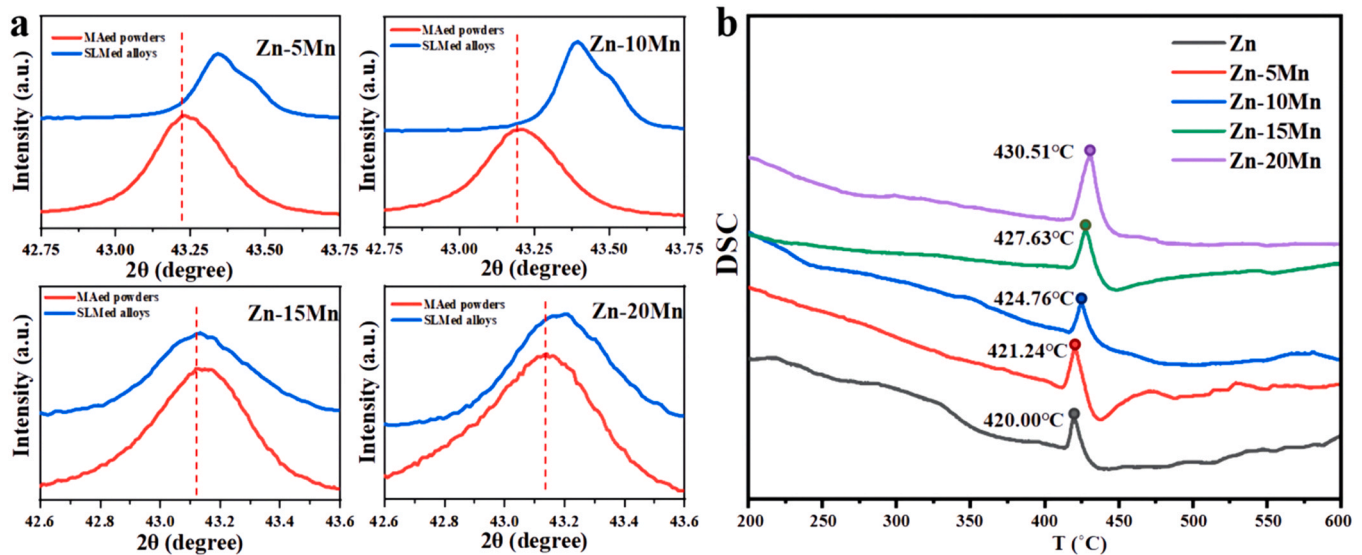


Fig. 5. (a) XRD patterns and (b) DSC curves of the SLMed Zn-xMn alloys.

small amount of Mn precipitation during SLM process. However the 0.23% and 0.52% shifting indicated that the majority of supersaturated solid solution structure formed in MA process was able to be retained during the SLM process. This phenomenon was remarkable, especially for Zn-15Mn and Zn-20Mn alloys. As could be seen in the XRD pattern, the major peak 101_{Zn} shifted to higher angle for only 0.02% (Zn-15Mn) and 0.13% (Zn-20Mn), compared with the corresponded MAed powders, which further proved the effect of SLM in retaining the supersaturated Zn-Mn solution structure formed during the MA process. On the one hand, the Zn-Mn supersaturated solid solution powders prepared by MA was in metastable state and would easily transformed into stable phase due to the thermal instability, which would lead to the precipitation of solute Mn from metastable solid solution during the consolidation process. On the other hand, as an eligible nonequilibrium rapid solidification technique, SLM was known for localized high heat input during short interaction time and rapid cooling rate up to $10^5 \sim 10^7$ K/s [36–39]. Studies have proved that this extremely fast cooling rate could keep the molten metal away from the immiscible zone in such immiscible metallic systems, hence inhibiting the precipitation of the solute element. The high cooling rate resulted from the fast scanning speed of SLM could be beneficial to retaining the supersaturated state and prepare non-equilibrium Zn-Mn alloys [44]. DSC analysis was conducted to further study the thermal properties of the SLMed pure Zn and Zn-xMn alloys, as shown in Fig. 5b. For all the alloys, there was only one endothermic peak at $\sim 420.0^\circ\text{C}$, which corresponded to the melting of Zn matrix. And as shown in the curves that when Mn content increased, endothermic peak of Zn-xMn alloys slightly shifted towards the direction of higher temperature, i.e. $\sim 424.8^\circ\text{C}$, $\sim 427.6^\circ\text{C}$ and $\sim 430.5^\circ\text{C}$ for SLMed Zn-10Mn, Zn-15Mn and Zn-20Mn alloys, respectively. The raised melting temperature might be ascribed to the higher melting point of Mn than Zn matrix and the refined microstructure by the combination process of MA and SLM.

In order to study the grain structures as well as the orientations, EBSD analysis was carried out on MAed and SLMed Zn-15Mn alloy (Fig. 6), as Zn-15Mn showed relatively better retention of the supersaturated Zn-Mn solid solution structure according to the XRD results (Fig. 5a). SLMed pure Zn and SLMed physical-mixed Zn-15Mn alloy were set as control groups. The misorientation distribution maps of grain boundary were shown in Fig. 6a–c, and the corresponding orientation maps were presented as inserts. Usually, the

grain boundaries were classified into two categories, including low-angle grain boundaries ($2^\circ \leq \text{LAGB} < 15^\circ$) and high-angle grain boundaries ($\text{HAGB} \geq 15^\circ$). It has been reported that HAGBs could effectively deflect the propagation or even inhibit the nucleation of microcracks, thereby contributing to the mechanical improvement [54]. The statistical data of misorientation angle distribution (Fig. 6d–f) revealed that the HAGBs increased from 37.4% for SLMed pure Zn and 36.6% for the physical-mixed and SLMed Zn-15Mn alloy to 54.8% for the MAed and SLMed Zn-15Mn alloy, indicating that Mn alloying via MA process could effectively increase the number of HAGBs and hence contribute to the mechanical performances of the alloys. Moreover, it should also be noted that pure Zn showed a refined grain size after physical-mixed and alloyed with Mn while the seemingly inhomogeneous grain sizes might be caused by the high Mn content. Interestingly, the MAed and SLMed Zn-15Mn alloy presented a much more refined and homogeneous grain sizes, which could be attributed to the homogeneous dissolved or dispersed Mn in the matrix by MA process. During MA process, the “microforging” effect of grinding balls on the powder and the continuous micro-strain would reduce the plasticity of powder and induce work hardening, resulting in the breaking of powder. At the same time, the crystal defect density in the powder would gradually increase with the increase of microstrain. When the defect density in the micro-strain band reached a critical value, the grain size would be further refined. In addition, the rapid solidification during the SLM process (with rapid cooling rate up to $10^5 \sim 10^7$ K/s [36–39]) could increase the degree of undercooling, which was also beneficial to hinder the grain growth. It was known that the refined grain size could lead to increased grain boundaries, the blocking and accumulation of dislocations, thereby contributing to mechanical enhancement.

3.2. Mechanical properties

Mechanical properties were considered as significant factor for bone implants, since sufficient structural support were needed at the implant position. The compressive performance of SLMed Zn-xMn alloys was measured and the results were displayed in Fig. 7a, with SLMed pure Zn as control. SLMed physical-mixed Zn-15Mn alloy was also set as control group, since Zn-15Mn alloy showed relatively better retention of the supersaturated Zn-Mn solid solution structure according to the XRD results (Fig. 5a), which indicated higher potential in mechanical enhancement. The physical-mixed Zn-15Mn

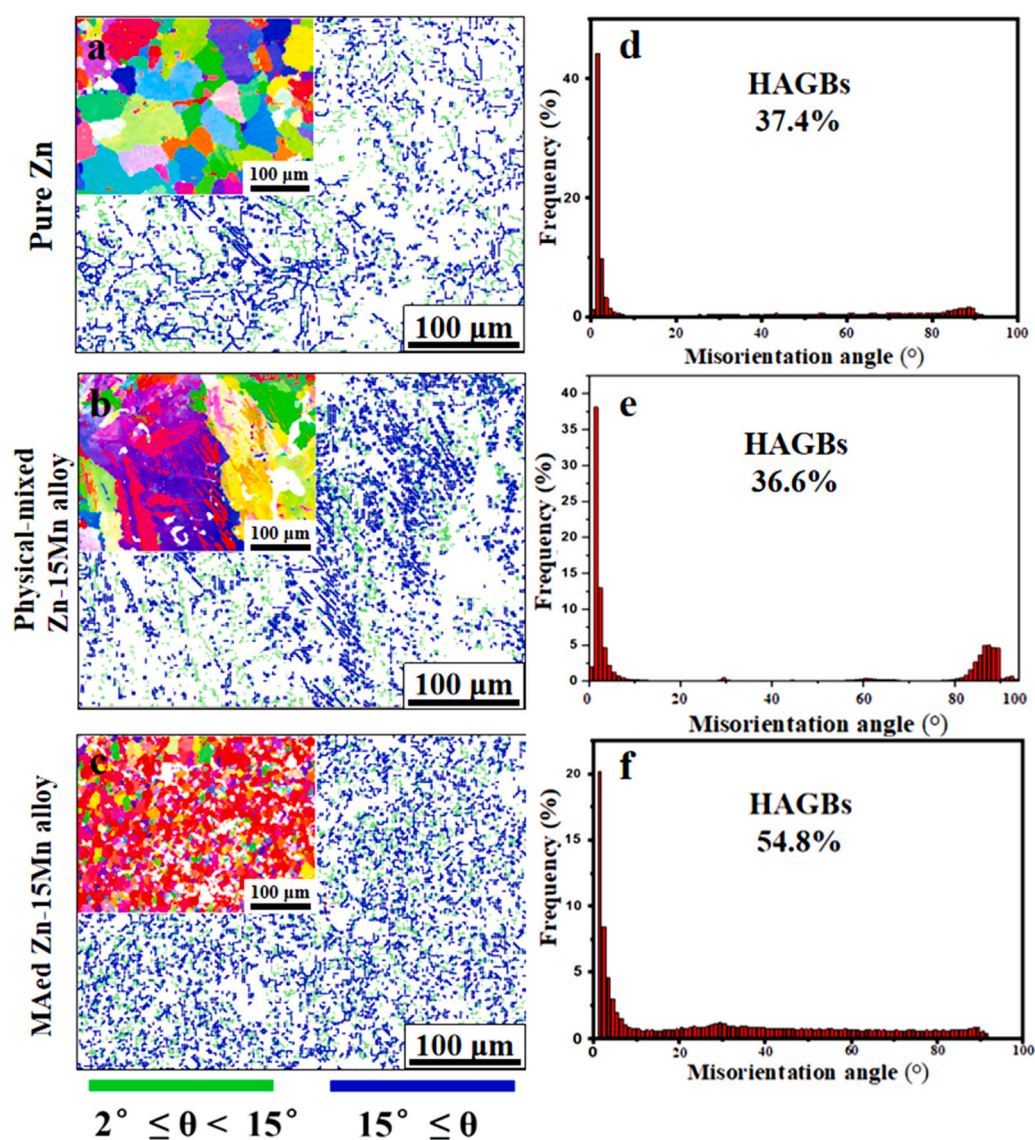


Fig. 6. (a-c) Grain boundary misorientation maps with orientation maps, and (d-f) misorientation distributions of the SLMed pure Zn, physical-mixed and SLMed Zn-15Mn alloy, and MAed and SLMed Zn-15Mn alloy, respectively.

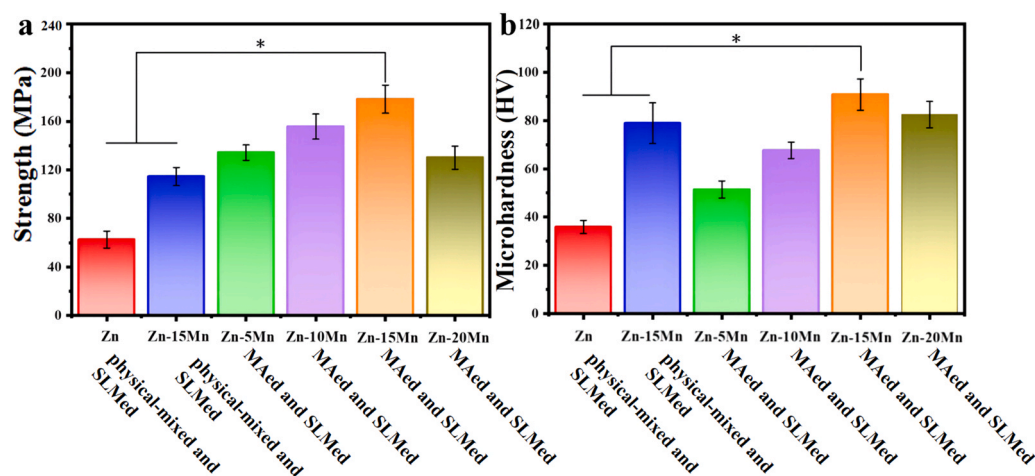


Fig. 7. The mechanical properties of the SLMed Zn-xMn alloys with SLMed pure Zn and Zn-15Mn alloy (physical-mixed) as control: (a) CYS, (b) microhardness.

alloy showed further enhanced strength with a compressive yield strengths (CYS) of 114 ± 7 MPa in comparison with pure Zn with CYS of 62 ± 8 MPa, which could only reach the bottom line of the strength requirements for human bone implants (110–180 MPa) [55–57]. Continues degradation of the Zn-Mn implant during its service would inevitably cause reduction on mechanical stability hence leading to premature failure of the scaffold with insufficient strength. In comparison with pure Zn and physical-mixed Zn-Mn alloy, the MAed and SLMed Zn-xMn alloys exhibited remarkably enhanced CYS. Specifically, the CYS increased with increasing Mn contents from 62 ± 8 MPa for pure Zn to 178 ± 11 MPa for the MAed and SLMed Zn-15Mn alloy. And microhardness results showed a similar trend (Fig. 7b) that pure Zn and physical-mixed Zn-15Mn alloy (without MA) presented a microhardness of 35 ± 2 HV and 79 ± 8 HV respectively, while that of MAed and SLMed Zn-15Mn alloy reached 90 ± 6 HV. These results demonstrated that the prepared Zn-xMn alloys were significantly strengthened and hardened, which were benefited from the supersaturated solution and the grain refinement by combination MA and SLM process. On the one hand, a large amount of solid solution of Mn atoms would cause serious lattice distortion of matrix (Zn) lattice, which would increase the resistance of dislocation movement and slip, thus increasing the strength and hardness of Zn-Mn alloys. On the other hand, grain boundary density was enhanced by grain refinement, followed by the promotion of the resistance to dislocation movement under load, resulting in mechanical enhancement. In addition, grain refinement also caused an escalation of the amount of grain boundaries, which was conducive to reduce stress concentration and impede crack propagation under external forces. Furthermore, according to the aforementioned EBSD analysis, the MAed and SLMed Zn-15Mn alloy exhibited a higher proportion of HAGBs in comparison with pure Zn and physical-mixed Zn-15Mn alloy, which could effectively inhibit the crack nucleation and hinder crack propagation. It was also worth noting that when the Mn content was further increased, the compressive strength and microhardness of Zn-20Mn alloy decreased to 130 ± 10 MPa and 81 ± 5 HV, respectively. This might be attributed to the too high content of Mn addition which was likely to form excessive aggregates or brittle intermetallic compounds, inevitably deteriorating the processability and mechanical properties of the alloys. In addition, the compressive yield strength of human bones is considered to be 110–180 MPa according to several studies [58–60]. The supersaturated solid solution Zn-Mn alloy prepared by MA and SLM combined process in this study presented a compressive yield strength of 178 MPa, which was within the strength requirements for most bone implants.

Fracture surfaces of SLMed pure Zn and Zn-xMn alloys were shown in Fig. 8. Pure Zn (Fig. 8a) exhibited a typical pattern of cleavage fracture mode, as a number of cleavage cracks and steps could be observed. These cleavage cracks interconnected with distorted grain boundaries and resulted in the formation of river-like strips, indicating the direction of the crack propagation. Compared with pure Zn, the fracture morphology of Zn-5Mn alloy (Fig. 8b) appeared with both partial cleavage planes and a few tiny dimples with finer fracture microstructure. This might be attributed to the extended solubility by the combined MA and SLM process. At the same time, a number of tearing ridges could be clearly observed, indicating a gradual change of the fracture mode from cleavage fracture into quasi-cleavage fracture. On the fracture surface of Zn-10Mn and Zn-15Mn alloys, many rough and tiny dimples were found (Fig. 8c,d), suggesting a more ductile fracture. Furthermore, when Mn content increased from 5 to 15 wt%, more fine dimples could be observed in SLMed Zn-xMn alloys. It meant that appropriate Mn alloying was beneficial for the transformation of fracture mode of pure Zn. However, different fracture morphology could be found in the SLMed Zn-20Mn alloy, in which many flat brittle facets were observed (Fig. 8e). This might be caused by the high Mn content in

Zn-20Mn alloy, which showed a comparable solubility of Mn to that of Zn-15Mn alloy according to the XRD results, and the excessive Mn was likely to exist in terms of large aggregates or brittle intermetallic compounds in the matrix.

3.3. Electrochemical performance

To assess the corrosion characteristics of the SLMed pure Zn and Zn-xMn alloys, electrochemical tests were carried out in SBF solution. Typical PDP curves were displayed in Fig. 9a. It showed noticeable reduction of E_{corr} from -0.920 V for pure Zn to -0.997 V for Zn-15Mn alloy. In addition, the I_{corr} of SLMed Zn-xMn alloys with different Mn contents ranged from $7.159 \mu\text{A}/\text{cm}^2$ to $15.871 \mu\text{A}/\text{cm}^2$. These results indicated a positive effect of appropriate Mn alloying by MA process on the biodegradability of Zn alloys. As could be seen from Fig. 9b that the radius of semicircle in Nyquist plots continuously decreased when Mn content increased. The Bode plots in Fig. 9c showed a similar downward trend of the alloy impedance, which usually represented the increase of degradation rate. On the one hand, due to the low corrosion potential of Mn, the supersaturated solute Mn would reduce the overall corrosion potential and promote the corrosion tendency of the Zn-xMn alloys. On the other hand, galvanic corrosion might occur between Zn matrix and dispersed Mn phases, which also accelerated the degradation of alloys. Furthermore, it could be seen from Fig. 9d that there were two peaks in the frequency-phase diagram of each alloy. In the high frequency region, Zn-15Mn alloy showed a larger frequency at the maximum phase angle, indicating a stronger capacitance behavior and hence faster degradation tendency [61]. Meanwhile, the largest phase angle of Zn-20Mn alloy indicated a more complete oxide film to impede the corrosion process, which was consistent with the Nyquist results. The equivalent circuit of Zn-xMn alloy was shown as an insert in Fig. 9. Since there were two time constants in the prepared Zn-xMn alloys, the equivalent circuit was based on double capacitance, where R_s was the solution resistance, which was determined by the concentration of electrolyte; R_{cpf} was the film resistance; R_{ct} was the charge transfer resistance; C_{dl} was the double-layer capacitance at the interface between the working electrode and electrolyte; Q_{cpf} was the corrosion product capacitance, which represented the degradation resistance of working electrode at low frequency band.

3.4. Cytocompatibility

Good cytocompatibility is necessary for bone implants to create an appropriate condition for cell proliferation and bone regrowth [62,63]. Cell fluorescence test and cell viability test are considered to be the most common methods in evaluating the biocompatibility of biodegradable implants [47,48]. Hence, in vitro cytocompatibility of the prepared alloys was assessed via co-culturing Zn-15Mn alloy extracts and MG-63 cells together, and pure Zn was selected as the control group (Fig. 10), as MAed and SLMed Zn-15Mn alloy showed the best mechanical enhancement. The cells were cultured in 10% Zn-15Mn extract for 1, 3 and 5 days followed by an observation by a fluorescence microscope. It could be found at the beginning of the experiment that the growth of MG-63 cells on Zn-15Mn alloy were slower than the ones on pure Zn. This could be attributed to the release of relatively higher concentrations Zn resulted from the accelerated degradation rate of Zn-15Mn alloy, as Zn could be cytotoxic to some extent at high concentrations [64,65]. It could be seen in Fig. 10a,b that the amount of MG-63 cells increased consistently with the extension of culture time, a certain amount of filopodia were also observed after cultured for 5 days, which evidenced the normal growth and proliferation of MG-63 cells. Moreover, CCK-8 tests were performed to assess the cytotoxicity of the prepared alloys. It was

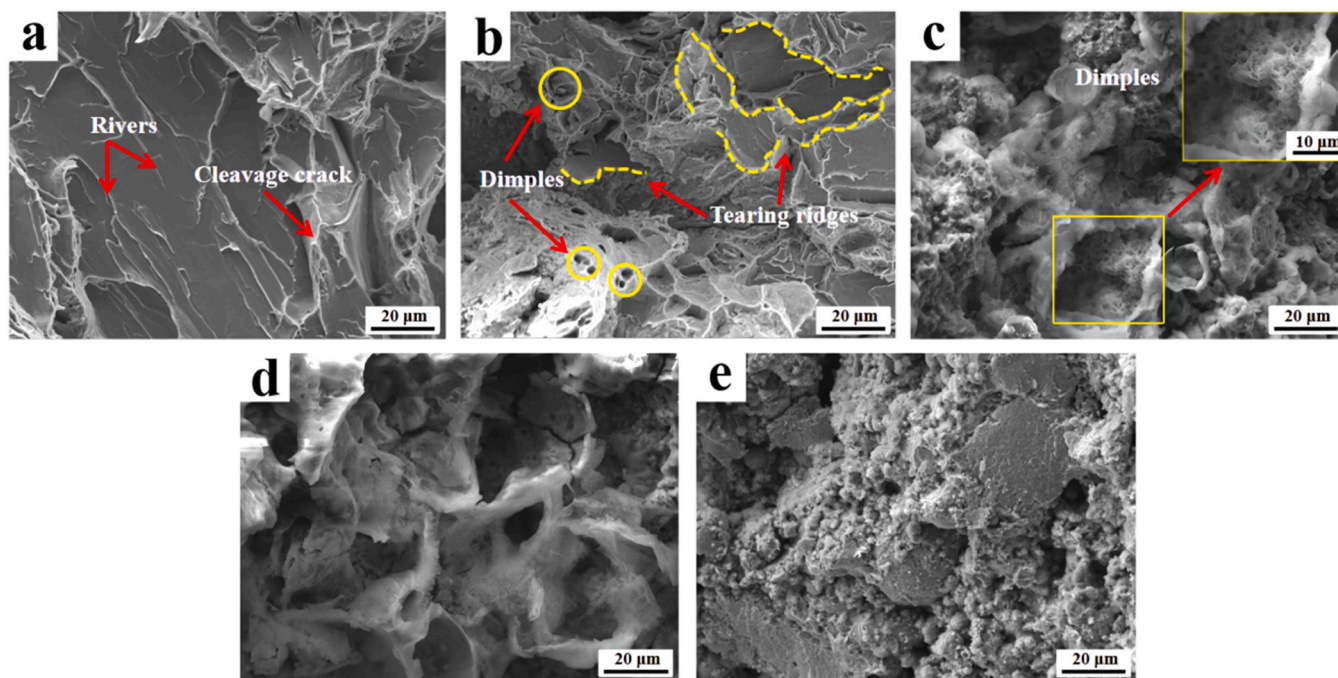


Fig. 8. SEM morphologies of the fracture surfaces of SLMed (a) pure Zn, (b) Zn-5Mn, (c) Zn-10Mn, (d) Zn-15Mn and (e) Zn-20Mn alloys.

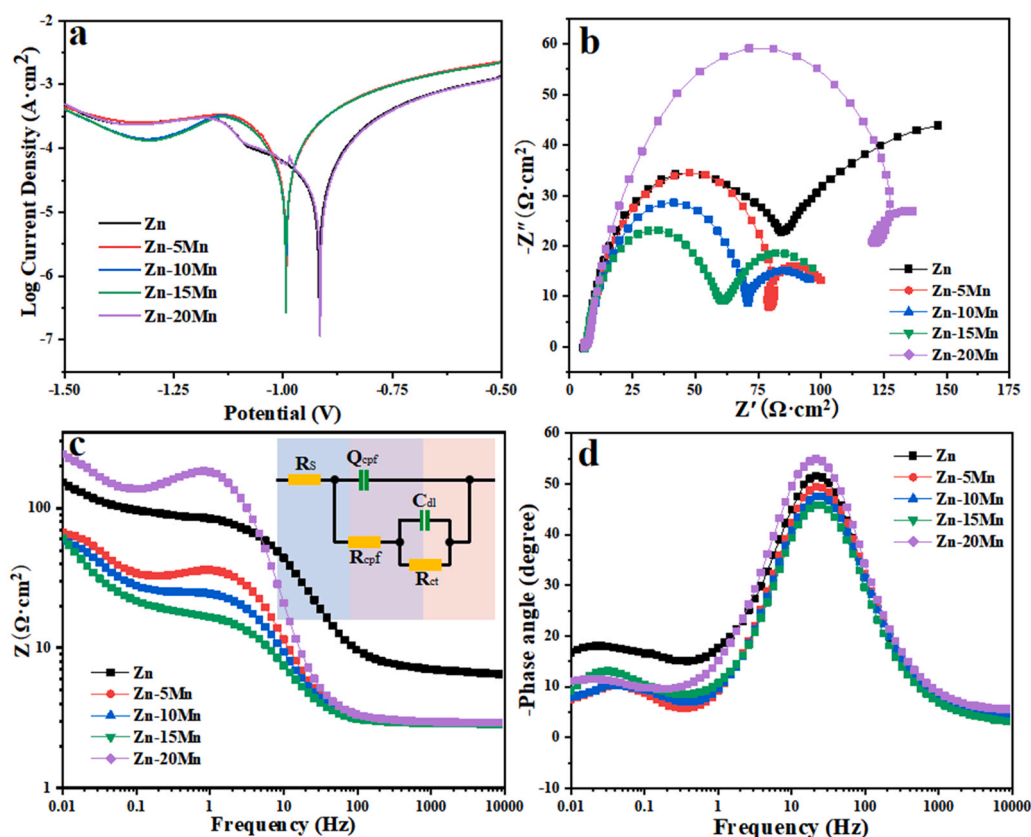


Fig. 9. Electrochemical results of the Zn-xMn alloys: (a) PDP curves, (b) Nyquist plots, (c) Bode plots of $\log |Z|$ vs. frequency, and (d) phase angles vs. frequency.

well recognized that a suitable dilution of the extract to 1:2–1:10 could better simulate the *in vivo* environment [66,67]. Therefore, the alloy extracts for CCK-8 tests were diluted to 50% and 10%, respectively. It could be seen in Fig. 10c that the OD value of the tested cells in 50% extract was about 0.1 after culturing for 5 days, which might

be caused by the relatively fast degradation and higher amount of released metallic ions. In comparison, the OD value of MG-63 cells in 10% extract was about 0.48 after culturing for 1 day and then increased to 2.75 after culturing for 5 days (Fig. 10d), indicating comparable cytocompatibility with pure Zn.

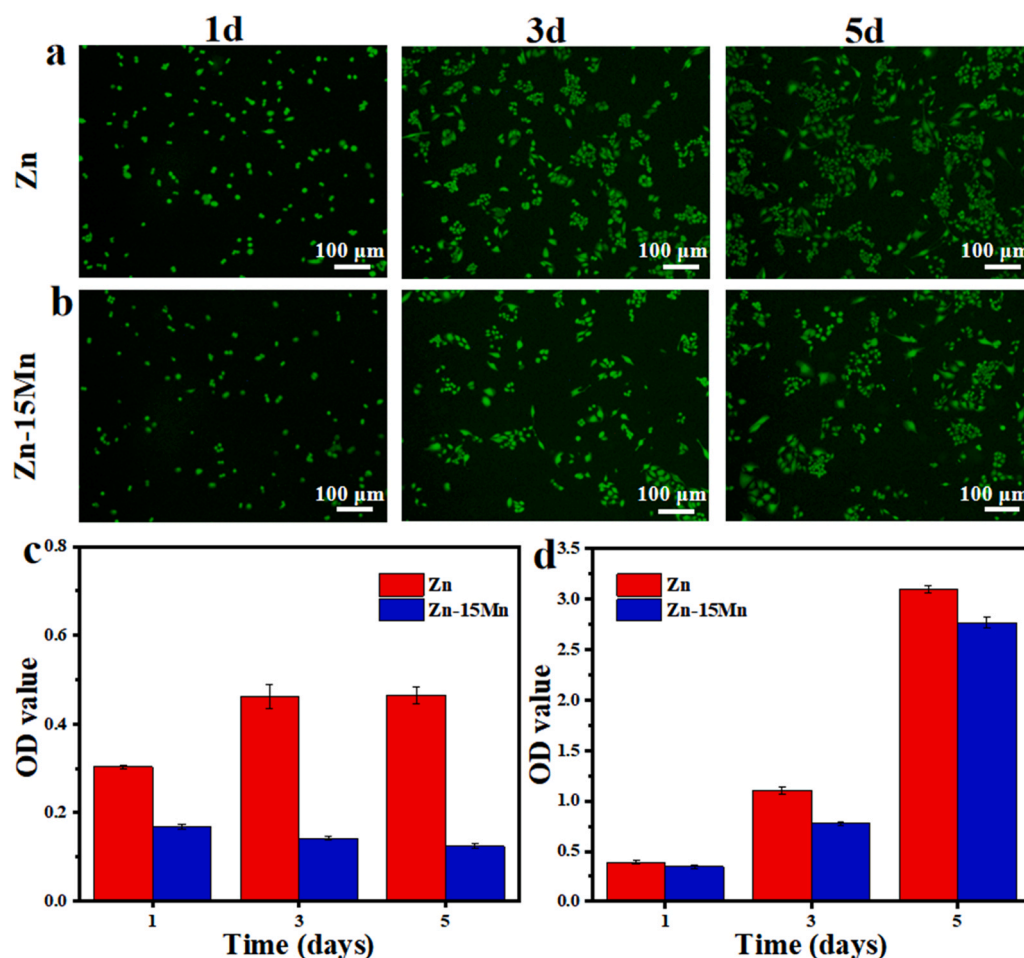


Fig. 10. (a, b) Fluorescence images of MG-63 cells cultured in 10% extract, and (c, d) OD values cultured in 50% and 10% extracts of Zn-15Mn alloys for 1, 3, and 5 days.

4. Conclusions

In this work, a combined MA and SLM process was used to prepare supersaturated solid solution Zn-xMn alloys, and the following conclusions could be drawn:

- (1) The MA process significantly extended the solid solubility of Mn in Zn powder and achieved supersaturated solid solution with maximum 3.6 wt% Mn dissolving in Zn lattice and the rest of Mn existed as homogeneously dispersed phases within the particles.
- (2) The SLM process effectively prevented the precipitation of the supersaturated solute Mn originated from MA process, thereby maintaining the metastable supersaturated structure in the Zn-Mn alloys.
- (3) Mn alloying via MA process also increased the HAGBs from 37.4% for pure Zn to 54.8% for Zn-15Mn alloy, as well as refined and homogeneous grain structure.
- (4) The Zn-15Mn alloy prepared by MA and SLM presented a compressive yield strength of 178 MPa which was 2.8 times and 1.6 times those of pure Zn and Zn-15Mn alloy without MA process, respectively, as well as a more ductile fracture, owing to solution strengthening and grain refinement.
- (5) The Zn-15Mn alloy presented accelerated corrosion caused by the supersaturated solution and galvanic corrosion, as well as comparable cytocompatibility with pure Zn.

These findings indicated that the combined process of MA and SLM may be an efficient strategy for preparing high-performance alloys with supersaturated solid solution and refined grain structure.

CRediT authorship contribution statement

Cijun Shuai: Conceptualization, Methodology, Investigation, Writing – review & editing. **Yang Zhao:** Investigation, Writing – original draft, Writing – review & editing. **Chuanzhi Li:** Resources, Investigation. **Youwen Deng:** Methodology, Formal analysis. **Zhenyu Zhao:** Formal analysis, Validation. **Chengde Gao:** Funding acquisition, Supervision, Project administration.

Data availability

Data will be made available on request.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This study was supported by the following funds: (1) The Natural Science Foundation of China [51935014, 52275395, 82072084]; (2) Hunan Provincial Natural Science Foundation of China [2020JJ3047]; (3) Central South University Innovation-Driven Research Programme [2023CXQD023]; (4) Jiangxi Provincial Natural Science Foundation of China [20224ACB204013]; (5) Technology Innovation Platform Project of Shenzhen Institute of Information Technology [PT2020E002]; (6) Guangdong Province Precision Manufacturing and

Intelligent Production Education Integration Innovation Platform [2022C]PT019]; (7) The Project of State Key Laboratory of High Performance Complex Manufacturing.

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